

INTRODUCTION

- Basic Goal Apply inputs to drive the state towards the origin
- Model-Based Paradigm Use a model of the system to design a control policy that maps states to inputs
- Issue Model unknown beforehand and must be estimated from a *finite number of noisy measurements*

Optimal Control Problem

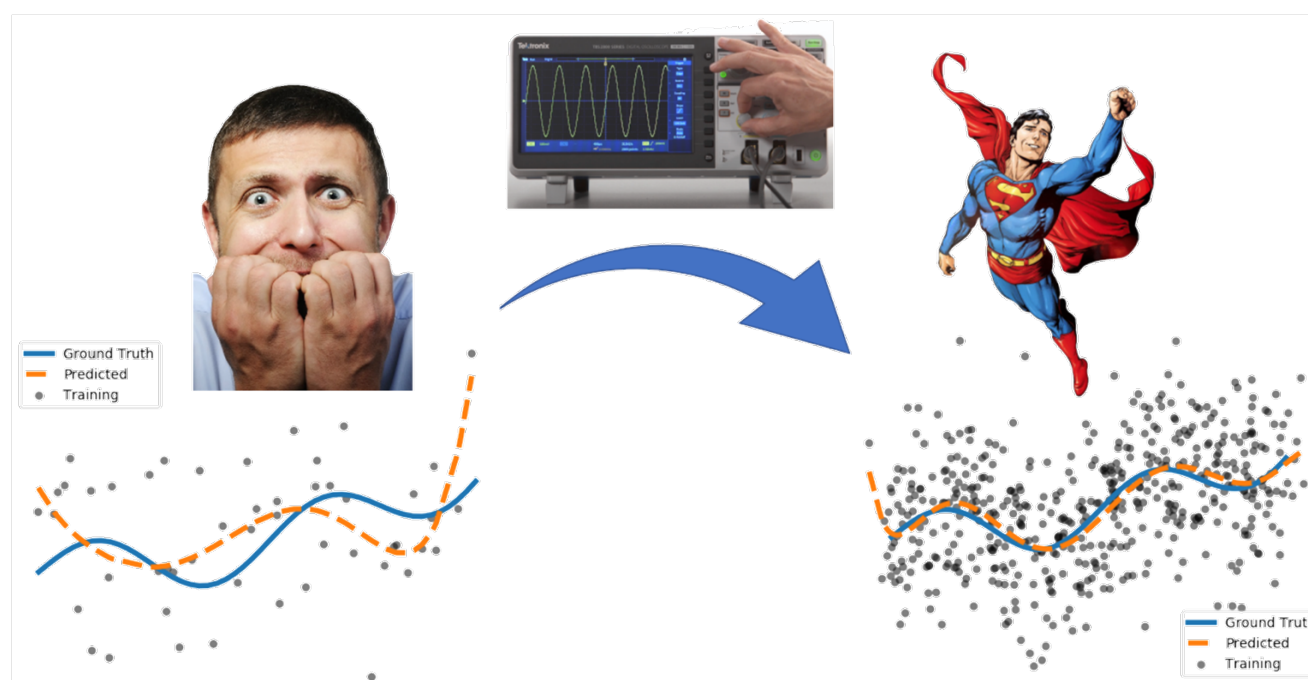
$$\min_{\pi} \mathbb{E} \left[\sum_{t=0}^T (x_t^T Q x_t + u_t^T R u_t - J^*) \right],$$

subject to $x_{t+1} = A x_t + B u_t + w_t,$

- $x_t \in \mathbb{R}^n$ is the system state, assumed known
- $u_t \in \mathbb{R}^m$ is the control input, assumed known and perfectly applied
- w_t is i.i.d. process noise with zero mean and covariance matrix W , assumed unknown
- (A, B) are system matrices, assumed unknown
- $Q \succeq 0$ and $R \succeq 0$ are cost matrices, assumed known
- $\pi = \{\pi_t\}_{t=0}^{T-1}$ are measurable history-dependent feedback policies
- J^* is the optimal infinite-horizon cost when (A, B) known

This motivates both *learning (adaptive)* control and *robust* control:

While we have significant model mis-specification we want to be robust to the uncertainty that exists, but as we gather more data we want to adapt the control to attain better performance.

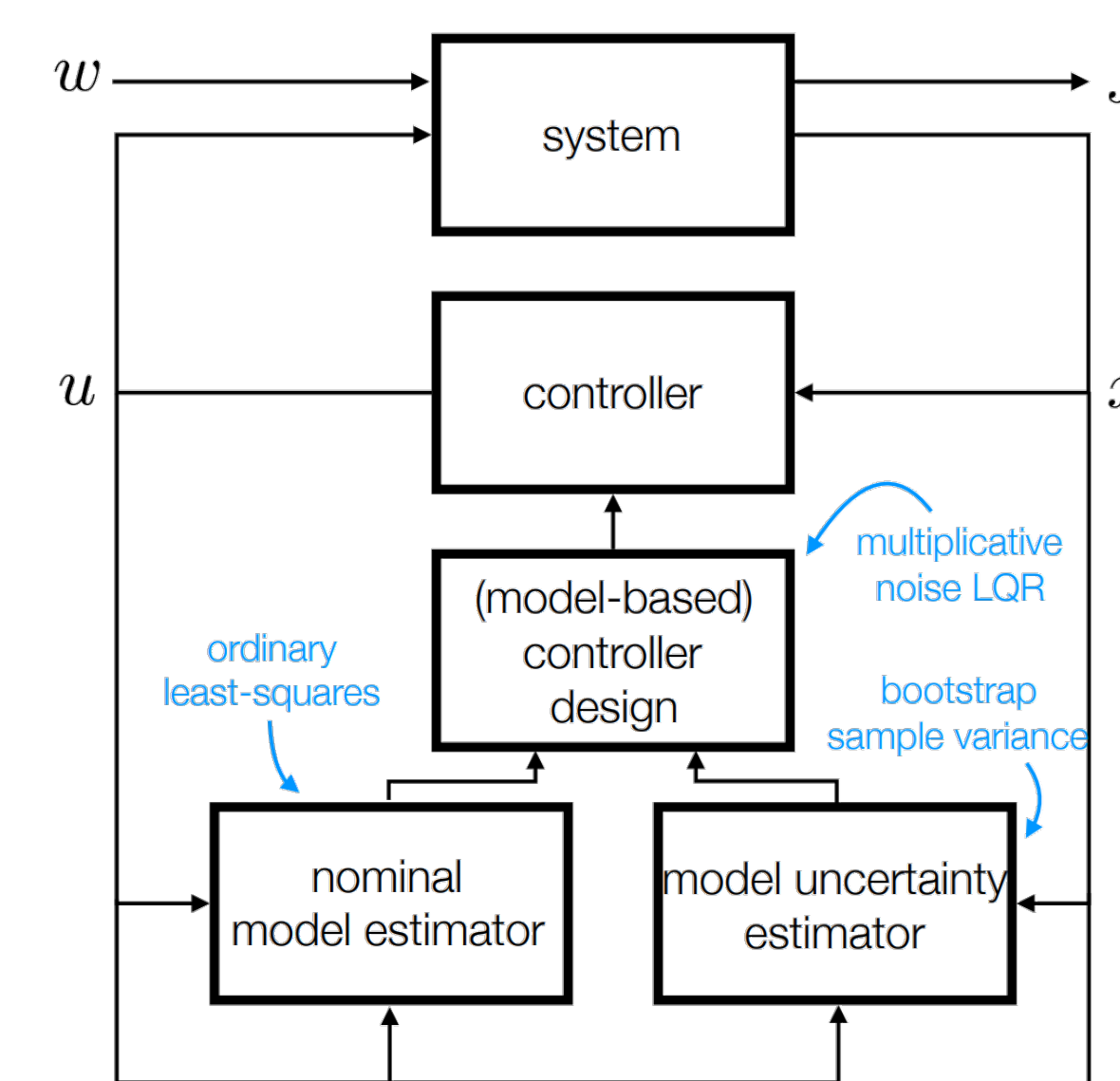


Certainty equivalent adaptive control can achieve asymptotic optimality and statistical efficiency with an *order* optimal rate [1]. However, this does not imply anything about non-asymptotic optimality or robustness.

PUTTING THE PIECES TOGETHER: ROBUST ADAPTIVE CONTROL

We propose a robust adaptive control architecture where the model uncertainty description and robust control design method both use stochastic uncertainty representations which align with each other:

- Least-squares nominal model estimator
- Bootstrap resampling to quantify non-asymptotic variance of the nominal model estimate
- Unconventional robust control design method using an optimal linear quadratic regulator (LQR) with multiplicative noise



Algorithm 1 Robust Adaptive Control

Input: exploration time t_{explore} , input excitation covariance U , number of bootstrap resamples N_b , model uncertainty scaling parameter γ , cost matrices $Q \succeq 0, R \succeq 0$

- $x_0 \sim \mathcal{N}(0, X_0)$
- for** $t = 0, \dots, T - 1$ **do**
- if** $t \leq t_{\text{explore}}$ **then**
- $u_t \sim \mathcal{N}(0, U)$
- $x_{t+1} = A x_t + B u_t + w_t$
- else**
- $(\hat{A}_t, \hat{B}_t) = \text{OrdinaryLeastSquares}(x_{0:t}, u_{0:t-1})$
- $(\hat{\Sigma}_A, \hat{\Sigma}_{B_t}) = \text{BootstrapModelVariance}(x_{0:t}, u_{0:t-1})$
- $\hat{K}_t = \text{MultiplicativeNoiseLQR}(\hat{A}_t, \hat{B}_t, Q, R, \gamma \hat{\Sigma}_A, \gamma \hat{\Sigma}_{B_t})$
- $e_t \sim \mathcal{N}(0, \|(\hat{\Sigma}_A, \hat{\Sigma}_{B_t})\| U)$
- $u_t = \hat{K}_t x_t + e_t$
- $x_{t+1} = A x_t + B u_t + w_t$
- end if**
- end for**

LINEAR QUADRATIC REGULATOR WITH MULTIPLICATIVE NOISE

$$\text{minimize}_{\pi \in \Pi} \mathbb{E}_{x_0, \{\bar{A}_t\}, \{\bar{B}_t\}} \sum_{t=0}^{\infty} (x_t^T Q x_t + u_t^T R u_t),$$

$$\text{subject to } x_{t+1} = (A + \bar{A}_t) x_t + (B + \bar{B}_t) u_t,$$

- $\bar{A}$ and $\bar{B}$ are iid zero-mean random matrices whose vectorizations have covariances Σ_A and Σ_B
- $\{\alpha_i, A_i\}_{i=1}^{n^2}$ and $\{\beta_j, B_j\}_{j=1}^{nm}$ are the eigenvalues and eigenvectors of Σ_A and Σ_B

The solution is linear state feedback, found from generalized algebraic Riccati equations:

$$P = Q + A^T P A + \sum_{i=1}^{n^2} \alpha_i A_i^T P A_i - A^T P B (R + B^T P B + \sum_{j=1}^{nm} \beta_j B_j^T P B_j)^{-1} B^T P A$$

$$\xrightarrow{\quad} K^* = -(R + B^T P B + \sum_{j=1}^{nm} \beta_j B_j^T P B_j)^{-1} B^T P A$$

It may be necessary to scale down the noise magnitudes to make the problem feasible:

Algorithm 3 Multiplicative Noise LQR

Input: Nominal model matrices A, B , cost matrices Q, R , multiplicative noise scaling γ and covariances $\hat{\Sigma}_A, \hat{\Sigma}_B$, bisection tolerance $\epsilon > 0$

- Find largest $c_\gamma \in [0, 1]$ via bisection such that there exists a feasible solution to $(P, K) = \text{GDARE}(A, B, Q, R, c_\gamma \gamma \Sigma_A, c_\gamma \gamma \Sigma_B)$

Output: Cost matrix P , gain matrix K

Resulting policy gives guaranteed stability robustness margins in a box region specified by the noise directions and magnitudes [2, 3]

Model uncertainty in various directions as multiplicative noises in the *same directions with the same magnitude*
Accomplished by estimating Σ_A and Σ_B using the bootstrap

LEAST-SQUARES SYSTEM IDENTIFICATION & BOOTSTRAPPING

Collect state-input trajectory $(x_{0:t}, u_{0:t})$ for t time steps, then form a *least-squares estimate (nominal system)*

$$(\hat{A}_t, \hat{B}_t) = \text{argmin}_{(A, B)} \left\{ \sum_{\tau=0}^{t-1} \|x_{\tau+1} - (A x_\tau + B u_\tau)\|_2^2 \right\} = X_t^\top Z_t (Z_t^\top Z_t)^{-1}$$

where $X_t^\top = [x_1 \ x_2 \ \dots \ x_t]$, $Z_t^\top = \begin{bmatrix} x_0 & x_1 & \dots & x_{t-1} \\ u_0 & u_1 & \dots & u_{t-1} \end{bmatrix}$

Use *bootstrapping* to estimate the covariance (uncertainty) in the nominal system while avoiding difficulties of dependence in time-series data:

- Draw random samples w/ replacement from dataset
- Construct lots of fake datasets of same size as original
- Compute system matrices for each fake dataset to obtain the bootstrap distribution of system matrices
- Use bootstrap distribution to estimate statistics e.g. covariance of the system matrices

Semiparametric bootstrap simulates nominal system with process noise sampled iid with replacement from residuals calculated with the nominal system

Less sensitive to assumptions about model and noise distribution

Better small sample performance when model is correctly specified

Nonparametric
bootstrap

Semi-parametric
bootstrap

Parametric
bootstrap

"The best of both worlds"

Algorithm 2 Semi-Parametric Bootstrap

Input: trajectory data $(x_{0:t}, u_{0:t-1})$, nominal model estimate $(\hat{A}_t, \hat{B}_t)$, residuals $\hat{w}_\tau = x_{\tau+1} - (\hat{A}_t x_\tau + \hat{B}_t u_\tau)$, $\tau = 0, \dots, t - 1$, number of bootstrap resamples N_b

- $\bar{x}_0 = x_0$
- $\bar{u}_{0:t-1} = u_{0:t-1}$
- for** $k = 1, \dots, N_b$ **do**
- Generate data $\bar{x}_{\tau+1} = \hat{A}_t \bar{x}_\tau + \hat{B}_t \bar{u}_\tau + \bar{w}_\tau$, $\tau = 0, \dots, t - 1$, where $\{\bar{w}_\tau\}_{\tau=0}^{t-1}$ is an iid resample with replacement from residuals $\{\hat{w}_\tau\}_{\tau=0}^{t-1}$
- $(\hat{A}_t^k, \hat{B}_t^k) = \text{argmin}_{(A, B)} \left\{ \sum_{\tau=0}^{t-1} \|\bar{x}_{\tau+1} - (A \bar{x}_\tau + B \bar{u}_\tau)\|_2^2 \right\}$
- end for**

Output: Bootstrap sample covariance $\hat{\Sigma}_{A_t} = \frac{1}{N_b-1} \sum_{k=1}^{N_b} \text{vec}(\hat{A}_t^k - \hat{A}_t) \text{vec}(\hat{A}_t^k - \hat{A}_t)^\top$
Bootstrap sample covariance $\hat{\Sigma}_{B_t} = \frac{1}{N_b-1} \sum_{k=1}^{N_b} \text{vec}(\hat{B}_t^k - \hat{B}_t) \text{vec}(\hat{B}_t^k - \hat{B}_t)^\top$

NUMERICAL EXPERIMENTS

For direct comparison, we train models using both control schemes on identical offline training data generated by

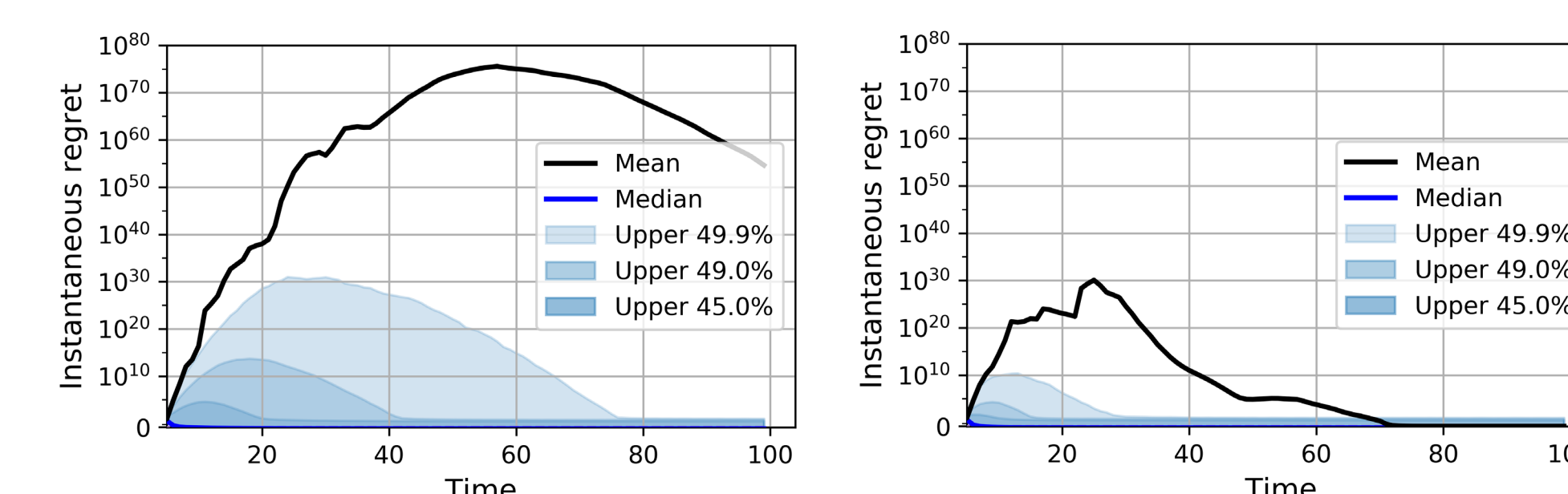
- Initializing the state at the origin
- Applying random Gaussian controls and additive process noise
- Simulating the evolution of the state

Instantaneous regret = adaptive - optimal control costs

We evaluated the algorithms on the scalar system

$$A = 1, B = 1, Q = 1, R = 0, W = 1$$

- Additive process noise is significant enough that many model estimates remain poor for many timesteps
- 100,000 independent Monte Carlo trials
- 100 bootstrap samples at each time step



Instantaneous regret using certainty equivalent (left) and robust adaptive (right) control

- Robust adaptive control achieves much lower expected regret (mean) and regret risk (upper quantiles)
- After several time steps, model estimates improve and difference becomes insignificant
- Heaviness of regret tails is shown by the massive difference between the mean and median

CONCLUSIONS

We proposed a robust adaptive control algorithm that uses the bootstrap to estimate model estimate covariances and uses these directly in a non-conventional multiplicative noise LQR robust control method.

Our approach provides a natural interface between two widely used and highly effective methods from statistics (bootstrap) and stochastic optimal control theory (LQR).

Numerical experiments significant outperformance on both expected regret and measures of regret risk.

Future work will go towards non-asymptotic robust stability and performance guarantees using mathematical tools from high-dimensional statistics.

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PAPER	https://arxiv.org/abs/2002.10069
CODE	https://github.com/TSummersLab/robust-adaptive-control-multinoise
CONTACT INFO	Benjamin Gravel benjamin.gravell@utdallas.edu Tyler Summers (PI) tyler.summers@utdallas.edu

[1] Horia Mania, Stephen Tu, and Benjamin Recht, "Certainty equivalence is efficient for linear quadratic control," in Advances in Neural Information Processing Systems, pages 10154–10164, 2019.
[2] Dennis Bernstein, "Robust Static and Dynamic Output-Feedback Stabilization: Deterministic and Stochastic Perspectives," in IEEE Transactions on Automatic Control, pages 1076–1084, Dec. 1987.
[3] Ben Gravel, Peyman Esfahani, and Tyler Summers, "Robust control design for linear systems via multiplicative noise," in 8th IFAC World Congress (to appear), arXiv preprint 2004.08019, 2020.